Public Transportation Delays - Data Model

Star Schema Structure :

FactTrips (Primary Table)

- Primary Key: trip_id
- Foreign Keys:
  - date → DimTime.date
  - hour → DimTime.hour
  - route_id → DimRoute.route_id
  - origin_station → DimStation.station_id
  - destination_station → DimStation.station_id
  - trip_status → DimTripStatus.status_id
- Measures :
  - actual_departure_delay_min
  - actual_arrival_delay_min
  - passenger_count
  - capacity_percentage
  - delayed
  - delayed_5min
  - on_time
  - peak_hour
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Dimension Tables

1. **DimRoute :** route_id (PK), transport_type, route_name
2. **DimStation :** station_id (PK), station_name, area
3. **DimTime :** date (PK), hour (PK), year, month, month_name, day, day_name, is_weekend
4. **DimTripStatus :** status_id (PK), status_name

Measures Definitions :

1. **Average Arrival Delay :**

Mean delay in minutes across all trips.

Avg Arrival Delay = AVERAGE(FactTrips[actual_arrival_delay_min])

2. **On-Time Arrival Rate :**

Percentage of trips arriving on time (delay ≤ 0 minutes).

On-Time Rate = DIVIDE(SUM(FactTrips[on_time]), COUNTROWS(FactTrips)) * 100

3. **Delayed >5min Rate :**

Percentage of trips delayed more than 5 minutes.

Delayed >5min % = DIVIDE(SUM(FactTrips[delayed_5min]), COUNTROWS(FactTrips)) * 100

4. **Total Passengers :**

Total passengers transported.

Total Passengers = SUM(FactTrips[passenger_count])

5. **Average Capacity Utilization :**

Mean capacity percentage across all trips

Avg Capacity % = AVERAGE(FactTrips[capacity_percentage])

6. **Delay Trend (Month-over-Month) :**

Percentage change in average delay compared to previous month

Delay MoM % = DIVIDE([Avg Arrival Delay]-CALCULATE([Avg Arrival Delay], PREVIOUSMONTH(DimTime[date])), CALCULATE([Avg Arrival Delay], PREVIOUSMONTH(DimTime[date]))) * 100

7. **Peak vs Off-Peak Delay Ratio :**

Ratio of average delay during peak hours to off-peak hours.

Peak Delay Ratio = DIVIDE(CALCULATE([Avg Arrival Delay], FactTrips[peak_hour]=1), CALCULATE([Avg Arrival Delay], FactTrips[peak_hour]=0))

8. **Route Delay Rank :**

Rank of routes based on average arrival delay.

Route Delay Rank = RANKX(ALL(DimRoute[route_name]), [Avg Arrival Delay], , DESC)

9. **Top 5 Delayed Routes :**

Identify five routes with highest average delays.

Top 5 Delayed Routes = TOPN(5, VALUES(DimRoute[route_name]), [Avg Arrival Delay])

10. **Origin Station Delay Count :**

Count of trips delayed >5min by origin station.

Origin Station Delay Count = CALCULATE(COUNTROWS(FactTrips), FactTrips[delayed_5min]=1, USERELATIONSHIP(FactTrips[origin_station], DimStation[station_id]))

11. **Origin Station Avg Delay :**

Average delay per origin station.

Origin Station Avg Delay = CALCULATE([Avg Arrival Delay], USERELATIONSHIP(FactTrips[origin_station], DimStation[station_id]))

12. **Cancellation Rate :**

Percentage of cancelled trips.

Cancellation Rate = DIVIDE(CALCULATE(COUNTROWS(FactTrips), DimTripStatus[status_name]="Cancelled"), COUNTROWS(FactTrips)) * 100